

3. Timescales in the Universe

1. Andromeda galaxy is roughly 2.2 Mly away. How far in the past do we see it?
2. How many meters away is Proxima Centauri, at a distance of 4.25 ly?
3. How is a telescope like a time machine?
4. If Pluto is 40 AU away from us, how many seconds would it take a radio signal sent from a spacecraft orbiting Pluto to reach Earth?
5. In a scale model of the solar system if we shrunk Jupiter (with a diameter of 142,984 km down by a factor of 5 billion times, how big would it be in cm?
6. Some timescales:
 - A. time for earth to orbit sun
 - B. time for moon to orbit earth
 - C. time for earth to spin once on its axis
 - D. time for moon to spin once on its axis
 - E. time for solar system to orbit Milky Way galaxy once
 - F. age of universe